

Assignment 2

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1 Stability Analysis

In order to determine a stable number of scenarios, i.e., such that the noise due to sampling demand from a probability distribution can be neglected, we perform two distinct analyses. The tests reported below are performed for a number of scenarios $N = [16, 32, 64, 128, 256, 512, 1024, 2048]$. The number of Monte Carlo runs was set to 50.

1.1 In-Sample Stability

Given two sets of scenarios $\mathbf{S}$ and $\mathbf{S}'$, with the same cardinality, we want that

$$\min_{x \in \mathcal{X}} f(x | \mathbf{S}) \approx \min_{x \in \mathcal{X}} f(x | \mathbf{S}') \quad (1)$$

The number of scenarios that satisfies this condition guarantees In-Sample stability. Adopting a 95% confidence interval, the stopping condition

$$0 \in \left[\bar{\phi}_N - z_{1-\alpha/2} \cdot \frac{\sigma_\phi}{\sqrt{N}}, \bar{\phi}_N + z_{1-\alpha/2} \cdot \frac{\sigma_\phi}{\sqrt{N}} \right] \quad (2)$$

is already reached for $N = 16$ scenarios. However, as can be seen from Figure 1, the confidence interval has a particularly large width. We observe a clear reduction at $N = 256$.

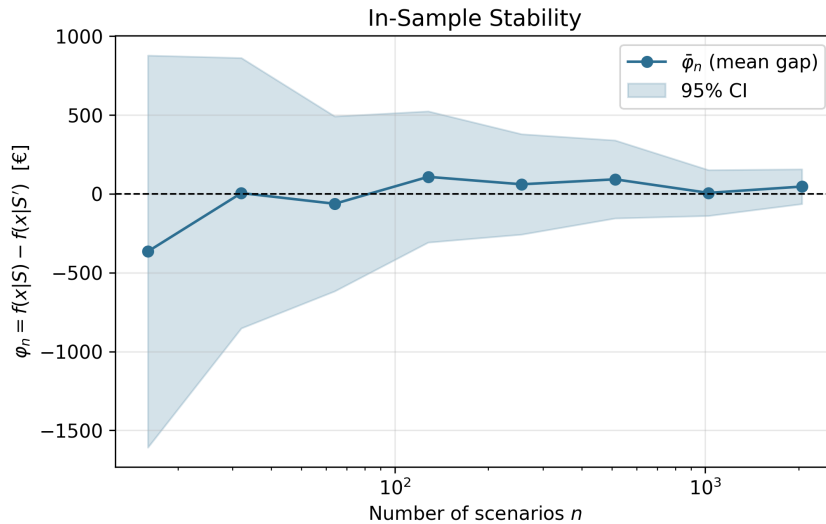


Figure 1: In-Sample stability as the number of scenarios varies

1.2 Out-of-Sample Stability

Given a set of scenarios $\mathbf{S}$ and a much larger one $\mathcal{E}$, we want that

$$f(x^* | \mathbf{S}) \approx f(x^* | \mathcal{E}) \quad (3)$$

where x^* is the solution to the two-stage problem. The number of scenarios that satisfies this condition guarantees Out-of-Sample stability. In this case, we set a percentage tolerance and require the stopping condition to be triggered when the difference between the objective function calculated with reference to $\mathbf{S}$ and in $\mathcal{E}$ is lower than the chosen tolerance. Adopting a 2% tolerance, stability is reached for $N = 256$

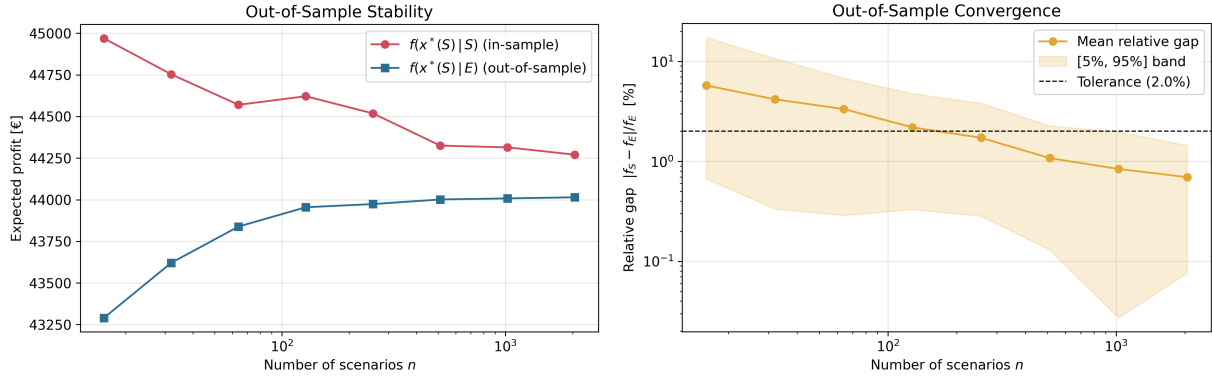


Figure 2: Out-of-Sample stability as the number of scenarios varies: convergence of the expected profit (left) and relative difference (right)

scenarios. However, a closer look at the quantiles of the distribution (Figure 2) makes it evident that this number needs to be increased to at least $N = 1024$.

In conclusion, in all subsequent analyses we adopt $N = 1024$. We specify that in the problem at hand, the low computational cost allows for a particularly conservative approach. However, in real-world size problems, a choice of this kind might not be convenient.

2 Metrics: VSS and EVPI

The results reported in Table 1 quantify the economic impact of uncertainty.

Metric	Value (€)
Wait-and-See (WS)	54,205.55
Recourse Problem (RP)	44,801.03
Expected Result of Expected Value (EEV)	43,878.50
Expected Value of Perfect Information (EVPI)	9,404.52
Value of the Stochastic Solution (VSS)	922.52

Table 1: Metrics ($N = 1024$)

The high value of the EVPI highlights the high cost imposed by the two-stage structure: since the first-stage decisions must be fixed before observing the demand, the loss compared to a perfect information scenario is substantial (about 21% of the RP).

On the other hand, although the VSS value is small in absolute terms, it still represents 2% of the RP: depending on the economic context, this figure could be more or less impactful.

In conclusion, the high EVPI value suggests that investments in forecasting tools or greater production flexibility (e.g., shorter lead times) could generate significant economic value.

3 Model Robustness

An incorrect estimation of the parameters produces the results in Figure 3. A positive perturbation indicates that the true parameter is greater than the estimated parameter, and vice versa in the case of a negative perturbation. We observe that:

- when a lower average demand occurs compared to the estimated one (and used to determine the first-stage variables), the expected profit collapses. On the contrary, a higher average demand compared to the estimated one produces benefits in terms of expected profit, though moderate when compared to the damage of overestimation.

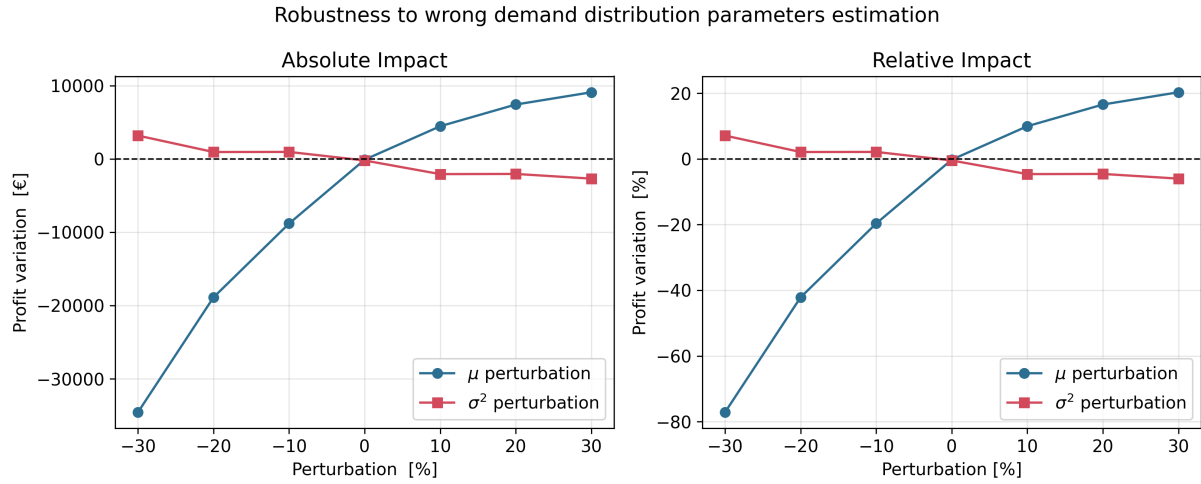


Figure 3: Effect of a perturbation on parameters μ and σ^2 on the profit ($N = 1024$). Absolute (left) and relative (right) variation.

The reason for this asymmetry is mainly attributable to the assumption of a salvage value equal to 0 for unsold goods.

- the model proves to be more robust to incorrect estimates on the variance than to incorrect estimates on the mean: the percentage impacts are much more contained (in the order of $\pm 6\%$), while potentially remaining relevant in a real-world context.